

એકેડેમિક કાર્યક્રમ તા. 17/02/2024

બાબત.....૦૬.....વિડિયો/પરિશિષ્ટ.....૦૨.....

Veer Narmad South Gujarat University**Bachelor of Business Administration Semester 3****MDC - Business Statistics**

Course Type	Multidisciplinary Course (MDC)
Course Title	<u>Business Statistics</u>
Credit	4
Teaching per Week	4 Hours
Review / Revision	June, 2024
Minimum weeks / Semester (Pedagogy)	15 (Lectures, Guest Lectures, Case Study, Presentations, Group Assignments)
Medium of Instruction	English
Purpose of Course	To provide students with basic understanding with the basic concepts of statistics and its applications in business
Course Objective	- To make students familiar with various measures of central tendency and dispersion. - To acquaint students with applicability of various business problems and their solution using statistics. - To explain the students with basics of data analysis.
Course Outcome	The students will be able to ... - understand evolution of Statistics, and how it contributes to solution of business-related problems with statistical procedures and techniques. - apply the concepts of business statistics in their research project work in subsequent semesters. - understand various distributions of the data and its applicability in business setup. - apply concepts of correlation and regression in decision making and forecasting. - get strong foundation for data and analytics.
Course Content	
Unit 1 Introduction to Business Statistics (Theory) (15%) - Brief History of Statistics, Meaning and Definitions of Business Statistics, Scope of Business Statistics, Applications of Business Statistics, - Types of Data: Qualitative and Quantitative, Primary and Secondary, Discrete and Continuous - Various types of graphs and charts used in statistics,	
Unit 2: Measures of Central Tendency & Dispersion (30%) Measures of Central Tendency: - Concept, Definitions, Advantages and Limitations, - Practical Problems on Mean, Median, Mode of Ungrouped data as well as Grouped data Measures of Dispersion:	

- Concept, Definitions, Advantages and Limitations,
- Range, Quartile Deviation, Mean Deviation, Standard Deviation (Merits, Demerits and Practical Examples of Grouped and Ungrouped Data)
- Coefficient of Variation and its uses, practical examples

Unit 3: Correlation and Regression

(25%)

Correlation

- Meaning, Definition, Types, Difference between Correlation and Causation, Properties of Correlation,
- Practical Examples on Karl Pearson's Method, Rank Correlation

Linear Regression

- Meaning, Definition, Uses, Limitations, Difference between correlation and Regression, Properties of Regression, Least Square method of Fitting Best line, Basic Understanding of Coefficient of determination (R^2)
- Practical Examples of Linear Regression

Unit 4: Testing of Hypothesis

(30%)

Conceptual Understanding (Theory):

- Hypotheses: Null and Alternative, Parameter & Statistic, Process of Hypothesis Testing, Sampling Distribution, Type I & II Errors, Confidence Intervals, Level of Significance, Acceptance Region,

T-Test:

- Test of Single Mean, Test of Difference between two means (Independent sample t), Paired sample t

Chi-Square Test:

- Test of Independence of Attributes (2x2 Table only),

Analysis of Variance:

- One Way & Two Way ANOVA

Suggested Readings:

1. Statistical Methods: S. P. Gupta, S. Chand & Co., New Delhi
2. Mathematical Statistics: S. C. Gupta, S. Chand & Co., New Delhi
3. Business Statistics: Ken Black, Wiley Publication
4. Business Statistics: J.K. Sharma, Vikas Publication House
5. Complete Business Statistics: Amir D. Aczel, Tata McGraw Hill
6. Business Statistics: S P Gupta & M P Gupta, Sultan Chand & Sons

Signature